

DSA CASE STUDY

ViewSync

Platform

Streaming Video Network Syndication Mesh

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Data Structures & Algorithms with C++

Problem Statement

Slow Catalog Classification

Organizing millions of video titles is inefficient and unscalable.

No Viewing State Rollback

Subscriber state changes cannot be tracked or reversed on error.

Random Task Execution

Profile processing tasks pile up with no guaranteed order.

Slow License Lookup

Finding a publisher trademark license takes too long.

No Delivery Optimization

There is no map of global licensing platforms to find lowest-cost delivery paths.

Objectives

- 1 Apply DSA to a real-world streaming application
- 2 Build a menu-driven C++ project using STL
- 3 Manage video catalogs efficiently with hash maps
- 4 Track viewing history with stack-based rollback
- 5 Process tasks in FIFO order using queue scheduling
- 6 Verify licenses using hashing techniques
- 7 Rank clips by viewer retention via sorting algorithms
- 8 Model distribution networks using graphs
- 9 Compute optimal delivery routes using Dijkstra's Algorithm
- 10 Balance encoding workloads using priority queues

System Architecture

1

Catalog Directory

2

Correction Log

3

Algorithm Pipeline

4

Registration Code

5

Clip Sorter

6

Syndication Grid

7

Delivery Transport

8

Frame Splitter

Data Structures & Algorithms Used

Catalog Directory

Vector, Unordered Map

Correction Log

Stack

Algorithm Pipeline

Queue

Registration Code

Unordered Map

Clip Sorter

STL Sort — $O(n \log n)$

Syndication Grid

Graph (Adjacency List)

Delivery Transport

Dijkstra's Algorithm

Frame Splitter

Priority Queue (Min-Heap)

Complexity Analysis

Operation	Time Complexity	Space Complexity
Add / Search Video	$O(1)$ Average	$O(n)$
Record / Undo Viewing State	$O(1)$	$O(n)$
Add / Process Pipeline Task	$O(1)$	$O(n)$
Verify License	$O(1)$ Average	$O(n)$
Sort Clips	$O(n \log n)$	$O(n)$
Add Graph Edge	$O(1)$	$O(V + E)$
Dijkstra's Algorithm	$O((V + E) \log V)$	$O(V)$
Frame Scheduling	$O(n \log p)$	$O(n)$

Module Implementation

Catalog Directory

Vectors + Unordered Maps for $O(1)$ average search by ID, genre, or tag.

Correction Log

Stack stores viewing state changes; `pop()` gives instant undo on error.

Algorithm Pipeline

Queue ensures FIFO processing of profile updates and recommendations.

Registration Code

Hash map keyed by license code allows constant-time region verification.

Clip Sorter

STL sort on retention values surfaces highest-performing clips first.

Syndication Grid

Adjacency list graph models platform relationships and licensing links.

Sample Execution Output

Video Catalog

ID	TITLE	GENRE	CATEGORY
101	Ocean Crimes	Thriller	Series
102	Galaxy Kids	Animation	Movie
103	Chef World	Reality	Show

License Verification

Code : LIC-NET-IND
Region : India
Result : License Verified ✓

Delivery Route

Source : Rights Holder
Dest : User Device
Cost : 25
Route : Holder → India CDN → Device

Frame Splitter

Processor 1 → Load: 14
Processor 2 → Load: 13
Processor 3 → Load: 10
Total Time : 14

Results & Findings

✓ Hash-based indexing enables $O(1)$ average content and license lookup

✓ Queue scheduling guarantees FIFO task execution for consistency

✓ Graph structures accurately model content distribution networks

✓ Priority queues distribute encoding workloads with minimal total time

✓ Stack operations provide instant rollback of viewing state changes

✓ STL sorting effectively ranks clips by viewer retention

✓ Dijkstra's Algorithm identifies optimal low-cost delivery routes

✓ All 8 DSA modules work cohesively to simulate a real streaming platform

Conclusion

The ViewSync Platform successfully demonstrates the practical application of Data Structures and Algorithms in a streaming platform environment.

- Vectors and Hash Maps for fast catalog management
- Stack-based Correction Log for reliable undo operations
- Queue-driven Pipeline for ordered task scheduling
- Graph + Dijkstra for optimal content delivery routing
- Priority Queue for balanced encoding workload distribution

[GitHub: github.com/sanikakangane/VIEWSYNC-PLATFORM](https://github.com/sanikakangane/VIEWSYNC-PLATFORM)